

TB Compressor

DETAILED ENGLISH USER MANUAL

A Peak/RMS dynamics processor with sidechain filtering, lookahead, stereo linking, saturation, circuit models, and oversampling.



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Host-visible endpoints documented: 21

1. Product purpose

A Peak/RMS dynamics processor with sidechain filtering, lookahead, stereo linking, saturation, circuit models, and oversampling.

Dynamic sources often need both level control and purposeful tone. A transparent compressor may control peaks but leave the source uninspiring; a colored compressor may add energy but obscure the dynamics decision. TB Compressor separates detection, gain reduction, sidechain focus, and saturation so each part can be set deliberately.

What result it creates

- Transparent leveling or assertive peak control
- Frequency-focused detector behavior
- Linked stereo image stability
- Optional harmonic color before final level trim

Signal flow

Input trim -> filtered peak or RMS detector with optional lookahead -> gain computer -> stereo link -> saturation/circuit stage with oversampling -> makeup and output

2. Complete feature guide

Peak and RMS detection

Peak mode reacts to instantaneous excursions and is suited to drums and protection. RMS mode follows average energy and is often smoother for vocals, bass, and buses.

Threshold, Ratio, and Knee

Threshold defines where compression begins, Ratio controls how strongly level above threshold is restrained, and Knee softens the transition. A wide knee is smoother; a narrow knee is more decisive.

Attack, Release, and Lookahead

Attack preserves or suppresses the front of a sound, Release determines recovery, and Lookahead allows earlier peak control at the cost of latency. Very short attack and release values can sound dense or distorted by design.

Sidechain filters

Low Cut prevents sub-bass from dominating detection. High Cut can reduce detector sensitivity to bright transients. These filters affect the detector, not the audible program path.

Stereo Link

Stereo Link applies a shared dynamics decision so the image does not tilt when one channel peaks. Disable only for intentionally independent channel movement.

Saturation system

Enable Saturation, select Clean, Soft, Hard, Punch, Warm, or Tape, then use Drive and Mix. The separate Circuit Model choices change response character. Oversampling reduces aliasing at higher CPU and latency cost.

Gain staging

Input changes how hard the detector and color stages are driven. Makeup restores intentional level after reduction; Output is the final trim. Compare at matched loudness.

3. Quick start

1. Set Input, Makeup, and Output to unity and disable Saturation.
2. Choose Peak or RMS detection.
3. Lower Threshold until gain reduction reaches the intended range.
4. Set Ratio and Knee for control character.
5. Tune Attack and Release in the context of the song.
6. Use sidechain filtering if lows or highs trigger too strongly.
7. Add saturation last and level-match with Output.

4. Practical workflows

Vocal leveling

1. Enable RMS.
2. Use a moderate ratio and soft knee.
3. Set Attack long enough to keep consonants clear.
4. Use a medium release that returns between phrases.
5. Apply light Warm or Tape saturation in parallel.

Expected result: The vocal stays forward and even without losing articulation.

Drum punch

1. Use Peak detection.
2. Set a slower attack to pass the initial hit.
3. Set release to recover before the next beat.
4. Use Punch saturation and oversampling if additional density is needed.

Expected result: Transients remain defined while the body becomes denser and more controlled.

Sidechain ducking

1. Route the key signal to the external sidechain bus.
2. Use sidechain filters to isolate the useful trigger range.
3. Choose fast attack and tempo-appropriate release.
4. Keep Stereo Link enabled for stable imaging.

Expected result: The target track moves out of the way predictably when the key signal arrives.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.

- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Lookahead and oversampling can change latency.
- Saturation Mix is not a substitute for compression Makeup.
- Heavy unlinked compression can move the stereo image.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 21 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Attack VST3 ID 1941037640	0.10 to 200.00	10.00	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Input VST3 ID 69820330	-24.00 to 24.00	0.00	Automatable	Sets the level entering the processor and therefore changes downstream drive or detection.
Knee VST3 ID 2311491	0.00 to 6.00	6.00	Automatable	Host-automatable control for Knee. Its full range and default are shown in this table; use it with the related feature section above.
Lookahead VST3 ID 1770736226	0.00 to 20.00	0.00	Automatable	Provides future-signal context to the associated dynamics or limiter stage and may add latency.
Makeup VST3 ID 119293193	0.00 to 24.00	0.00	Automatable	Sets gain for the named input, output, band, or parallel send/return path.
Output VST3 ID 195300609	-24.00 to 24.00	0.00	Automatable	Sets or limits the final output level after the plug-in's main processing.
Oversampling VST3 ID 1589895355	1x (Off), 2x, 4x, 8x	1x (Off)	Automatable	Selects the internal oversampling factor, trading more CPU/latency for cleaner non-linear processing.
Program VST3 ID 1886548852	Clean Control, Vocal Leveler, Drum Punch, Bus Glue, Sidechain Duck	Clean Control	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Ratio VST3 ID 77748203	1.00 to 20.00	2.00	Automatable	Sets the strength of the associated dynamics response.
Release VST3 ID 1808577511	5.00 to 1000.00	100.00	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.
RMS VST3 ID 81272	Off, On	Off	Automatable	Host-automatable control for RMS. Its full range and default are shown in this table; use it with the related feature section above.
Saturation VST3 ID 254601170	Clean, Soft, Hard, Punch, Warm, Tape	Clean	Automatable	Sets the strength of the associated dynamics response.
Saturation Drive VST3 ID 1944639409	0.00 to 4.00	1.00	Automatable	Sets the strength of the associated dynamics response.
Saturation Enabled VST3 ID 1252710312	Off, On	Off	Automatable	Sets the strength of the associated dynamics response.
Saturation Mix VST3 ID 442254915	0.00 to 1.00	1.00	Automatable	Sets the balance between the original signal and the complete processed path.
Saturator Circuit Model VST3 ID 1341934291	Legacy, XXL, LX-2X, FET-X76, FX-67X	Legacy	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Sidechain High Cut VST3 ID 84552468	100 to 20000	20000	Automatable	Reduces high-frequency content or shortens high-frequency decay in the associated path.
Sidechain Low Cut VST3 ID 1315373992	20 to 10000	20	Automatable	Reduces low-frequency content in the associated audio or detector path.
Stereo Link VST3 ID 2030866401	Off, On	On	Automatable	Links channel control so level changes do not pull the stereo image toward one side.
Threshold VST3 ID 1241117771	-60.00 to 0.00	-18.00	Automatable	Sets the level at which the associated detector, dynamics stage, clipper, or limiter begins to act.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible

controls are documented.